

2023 Summer Qual

III.

$$\text{Let } G_1 \text{ and } G_2 \text{ s.t. } AG_1A^T A = A, \\ AG_2A^T A = A$$

$$0 = (AG_1A^T A - AG_2A^T A)(G_1^T A^T - G_2^T A^T) \\ = \underbrace{(AG_1A^T - AG_2A^T)}_{= B^T} (AG_1A^T - AG_2A^T)^T$$

$$= B^T B$$

$$\Rightarrow B = 0$$

$$\Rightarrow AG_1A^T - AG_2A^T = 0 \rightsquigarrow AG_1A^T = AG_2A^T$$

So, $A(ATA)^+A^T$ is invariant to the choice of $(ATA)^+$.

2

$$\text{mgf}_{y|A y}(\pm) = \mathbb{E}[e^{\pm y^T A y}]$$

$$= \int_{-\infty}^{\infty} \frac{1}{(\det P)^{1/2} |\Sigma|^{1/2}} e^{\pm y^T A y} e^{-\frac{1}{2}(y-\mu)^T \Sigma^{-1}(y-\mu)} dy$$

$$= \int_{-\infty}^{\infty} \frac{1}{(\det P)^{1/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}[y^T(I-2\pm A\Sigma)\Sigma^{-1}y - 2y^T\Sigma^{-1}\mu + \mu^T\Sigma^{-1}\mu]} dy$$

For small $\pm$, $\mathbb{E}(I-2\pm A\Sigma)^{-1}$

$$\theta^T = \mu^T(I-2\pm A\Sigma)^{-1}, \quad \bar{V}^{-1} = (I-2\pm A\Sigma)\Sigma^{-1}$$

$$\text{mgf}_{y|A y}(\pm) = \int_{-\infty}^{\infty} \frac{1}{(\det P)^{1/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(y-\theta)^T \bar{V}^{-1}(y-\theta)} \cdot e^{-\frac{1}{2}[\mu^T \Sigma^{-1} \mu - \theta^T \bar{V}^{-1} \theta]} dy$$

$$= |I-2\pm A\Sigma|^{1/2} e^{-\frac{1}{2} \mu^T \underbrace{\Sigma^{-1}(I-(I-2\pm A\Sigma)^{-1})I}_{\text{scalar}} \mu}$$

$$= |I-2\pm A\Sigma|^{1/2} \exp\left\{-\frac{1}{2} \mu^T [(I-(I-2\pm A\Sigma)^{-1})\Sigma^{-1}] \mu\right\}$$

□

[3]

① $C\hat{\beta} = C^T(X^T X)^{-1}X^T Y$ is a linear combination of Y

② $E[C\hat{\beta}] = C\beta$: unbiased.

③ Let $\tilde{\beta} = d^T Y$ be any linear unbiased estimator of $C\beta$

$$E[\tilde{\beta}] = d^T X \beta = C\beta \Rightarrow d^T X = C^T$$

$$\begin{aligned} 0 &\leq \text{Var}(\tilde{\beta} - C\hat{\beta}) = \text{Var}((d^T - C^T(X^T X)^{-1}X^T)Y) \\ &= (d^T - C^T(X^T X)^{-1}X^T) \sigma^2 I (d - X(X^T X)^{-1}C) \\ &= \sigma^2 (d^T d - C^T(X^T X)^{-1}C - C^T(X^T X)^{-1}C + C^T(X^T X)^{-1}C) \\ &= \text{Var}(\tilde{\beta}) - \text{Var}(C\hat{\beta}) \end{aligned}$$

$\Rightarrow \text{Var}(C\hat{\beta}) \leq \text{Var}(\tilde{\beta})$: minimum variance.

$\therefore C\hat{\beta}$ is BLUE of $C\beta$

□

□

$$(a) Y = X_1 \beta_1 + \varepsilon \quad \text{RM}$$

$$Y = X_1 \beta_1 + X_2 \beta_2 + \varepsilon \quad \text{FM}$$

$$R_{\text{RM}}^2 = \frac{Y^T (H_{X_1} - H_1) Y}{Y^T (I - H_1) Y}, \quad R_{\text{FM}}^2 = \frac{Y^T (H_X - H_1) Y}{Y^T (I - H_1) Y}$$

$$H_X = H_{X_1} + H_{X_{2+}}, \quad X_{2+} = (I - H_{X_1}) X_2$$

$$X = (X_1, X_{2+})$$

$$X(X^T X)^+ X^T = \begin{pmatrix} X_1 & X_{2+} \end{pmatrix} \begin{pmatrix} X_1^T X_1 & 0 \\ 0 & X_{2+}^T X_{2+} \end{pmatrix} \begin{pmatrix} X_1^T \\ X_{2+}^T \end{pmatrix}$$

$$= H_{X_1} + H_{X_{2+}}$$

$$R_{\text{FM}}^2 - R_{\text{RM}}^2 \Leftrightarrow Y^T (H_X - H_1) Y - Y^T (H_{X_1} - H_1) Y$$

$$= Y^T (H_X - H_{X_1}) Y$$

$$= Y^T \underbrace{H_{X_{2+}}}_{\substack{\text{sym \& idemp} \\ \Rightarrow \text{p.s.d}}} Y \geq 0$$

□

$$(b) X_* = XA, \quad A \text{ is full rank.}$$

$$H_{X_*} = X_* (X_*^T X_*)^{-1} X_*^T = XA (A^T X^T X A)^{-1} A^T X$$

$$= XA A^{-1} (X^T X)^{-1} (A^T)^{-1} A^T X$$

$$= X (X^T X)^{-1} X^T = H_X$$

$$\frac{Y^T(H_X - H_1)Y}{Y^T(I - H_1)Y} = \frac{Y^T(H_X - H_1)Y}{Y^T(I - H_1)Y} = R^2$$

$$Z = CY$$

$$\frac{Z^T(H_X - H_1)Z}{Z^T(I - H_1)Z} = \frac{\cancel{c^2} Y^T(H_X - H_1)Y}{\cancel{c^2} Y^T(I - H_1)Y} = R^2$$

□

$$\begin{aligned} (c) \quad R^2 &= \frac{SSR}{ST} = \frac{SSR}{SSR + SSE} \quad \& \quad (H_X + H_1)(I - H_X) \\ &= H_X - H_X - H_1 + H_1 \\ &= 0 \\ &= \frac{\frac{SSR}{\cancel{c^2}}}{\frac{SSR}{\cancel{c^2}} + \frac{SSE}{\cancel{c^2}}} = \frac{q^2(p)}{q^2(p) + q^2(n-p-1)} = 0 \end{aligned}$$

$$= \frac{\Gamma(\frac{p}{2}, 2)}{\Gamma(\frac{p}{2}, 2) + \Gamma(\frac{n-p-1}{2}, 2)} \sim \text{beta}(\frac{p}{2}, \frac{n-p-1}{2})$$

□

5.

$$(a) (y - X\beta)^T (y - X\beta) + \beta^T \lambda \beta$$

$$\frac{\partial (y - X\beta)^T (y - X\beta) + \beta^T \lambda \beta}{\partial \beta}$$

$$= -2X^T(y - X\beta) + 2\lambda\beta = 0$$

$$\Rightarrow X^T y - X^T X \beta - \lambda \beta = 0$$

$$\Rightarrow (I + \lambda) \beta = X^T y$$

$$\Rightarrow \hat{\beta}^R = (I + \lambda)^{-1} X^T y$$

$$\begin{aligned} \text{Var}(\hat{\beta}^R) &= \sigma^2 (I + \lambda)^{-1} (I + \lambda)^{-1} \\ &= \sigma^2 (I + \lambda)^{-2} \end{aligned}$$

$$\text{tr}(\text{Var}(\hat{\beta}^R)) = \sum_{i=1}^p \frac{1}{(1 + \lambda_i)^2} \sigma^2$$

$$\text{Bias}(\hat{\beta}^R) = (I + \lambda)^{-1} \beta - \beta \Rightarrow \frac{1}{1 + \lambda} \beta - \beta = \frac{\lambda}{1 + \lambda} \beta$$

$$\text{Bias}(\hat{\beta}^R)^T \text{Bias}(\hat{\beta}^R) = \sum_{i=1}^p \left(\frac{\lambda_i}{1 + \lambda_i} \right)^2 \beta_i^2$$

$$\text{SMSE} = \sum_{i=1}^p \frac{1}{(1 + \lambda_i)^2} \left[\sigma^2 + \lambda_i^2 \beta_i^2 \right]$$

$$(b) \frac{\partial \text{SMSE}}{\partial \lambda_i} = \frac{2\lambda_i \beta_i^2 (1 + \lambda_i)^2 - (\sigma^2 + \lambda_i^2 \beta_i^2) 2(1 + \lambda_i)}{(1 + \lambda_i)^4}$$

$$= 0$$

$$\Rightarrow (1 + \lambda_i) \{ 2\lambda_i^2 \beta_i^2 + 2\lambda_i \beta_i^2 - 2\sigma^2 - 2\lambda_i^3 \beta_i^2 \}$$

$$= 0$$

$$\Rightarrow \lambda_i = \frac{\sigma^2}{\beta_i^2}$$

$$\text{SMSE} = \sum_{i=1}^p \frac{1}{(1 + \frac{\sigma^2}{\beta_i^2})^2} \left\{ \sigma^2 + \frac{\sigma^4}{\beta_i^2} \right\}$$

$$= \sum_{i=1}^p \frac{\sigma^2 \beta_i^2}{\beta_i^2 + \sigma^2}$$

□

$$\boxed{6}. S_x(u) = \begin{cases} 1 & , u \geq x \\ 0 & , u < x \end{cases}$$

$$x > Q(p),$$

$$F_{\pm}(y) = (1 \pm t)F(y) + tS_x$$

$$= (1 \pm t)(1 - e^{-y})$$

$$= p$$

$$\frac{p}{1 \pm t} = 1 - e^{-y} \Rightarrow e^{-y} = 1 - \frac{p}{1 \pm t}$$

$$\Rightarrow y = -\log\left(1 - \frac{p}{1 \pm t}\right)$$

$$IF(x; T, F) = \frac{d}{dt} - \log\left(1 - \frac{p}{1 \pm t}\right) \Big|_{t=0}$$

$$= \frac{1}{1 - \frac{p}{1 \pm t}} \cdot \frac{p}{(1 \pm t)^2} \Big|_{t=0}$$

$$p = 1 - e^{-y}$$

$$\Rightarrow y = -\log(1-p)$$

$$= \frac{p}{1-p} \quad , \quad x > Q(p) = -\log(1-p)$$

$$x \leq Q(p),$$

$$F_{\pm}(y) = (1 \pm t)F(y) + tS_x$$

$$= (1 \pm t)(1 - e^{-y}) + t$$

$$= p$$

$$\Rightarrow 1 - z = (1 - z)(1 - e^{-y})$$

$$\Rightarrow \frac{1-p}{1-z} = e^{-y}$$

$$\Rightarrow y = -\log\left(\frac{1-p}{1-z}\right)$$

$$\mathbb{E}F(d; T, F) = \frac{d}{dt} \left(-\log\left(\frac{1-p}{1-z}\right) \right) \Big|_{z=0}$$

$$= \frac{-1}{1-p} \frac{\frac{1-p}{1-z}}{\frac{1-p}{1-z}} \Big|_{z=0}$$

$$= -1, d \leq Q(p) = -\log(1-p)$$

$$\circ \circ \mathbb{E}F(d; T, F) = \begin{cases} -1 & d \leq -\log(1-p) \\ \frac{p}{1-p} & d > -\log(1-p) \end{cases} \quad \square$$